Table 6.3: Some common substrates used in the industrial fermentation process:

Carbon source	
Molasses	It is a byproduct of sugar industry. It is a cheap source of carbohydrates It also contains nitrogenous substances, vitamins, trace elements. (Example:) sugar cane, beetroot molasses
Malt extract	It is an aqueous extract of malted barley
starch, dextrin cellulose	They can be metabolized by microorganism. They used for the industrial production of alcohol
Whey	It is a byproduct of dairy industry used in the production of alcohol, SCP, vitamin B12, lactic acid, gibberllic acid
Methanol ethanol	Methanol in the cheapest substrate. It is utilized only by few bacteria yeast. Methanol is used for SCP. Ethanol is used for acetic acid production
Hydro molasses	It is a byproduct in glucose production from corn
Sulphate waste liquor	It is a spent sulfite liquor form the paper pulping industry. It is used in the production of ethanol by *Saccharomyces cerevisiae*, and in the growth of *Torula utilis* as a feed
Nitrogen sources	
Inorganic: Ammonium salts and ammonia	It is a cheap source of nitrogen
Urea (Organic)	It is a good and cheap source of organic source
Corn steep liquor (Organic)	It is formed during starch production from corn. It is rich in several amino acids
Yeast extract	It is rich in amino acids, peptides vitamins
Soy meal	It is a left out residue on preparing soybean oil from soybean seeds. It is used in antibiotic production
Peptones	The proteins hydrolysates are called as peptones. The source of peptones includes meat, cotton seeds and sunflower seeds

pure form. Examples: Phenyl acetic acid or phenylacetamide added as side chain precursors in penicillin production.

6.5.3 Large Scale Production

Basic Steps of Industrial Fermentation

Successful development of a fermentation process and fermentors requires major contributions from a wide range of other disciplines, particularly biochemistry, genetics, molecular biology, chemistry, chemical engineering and process engineering, mathematics and computer technology. A typical operation involves both upstream processing (USP) and downstream processing (DSP) stages (Figure 6.7).

6.5.4 Upstream Processing

It is the first step in which biomolecules like bacteria or other cells are grown in a fermentor. Upstream processing involves inoculation development, scale up, medium preparation and sterilization of media and fermentation process.

TN_GOVT_XII_Micro_Biology_CH06.indd 97 27-02-2019 14:46:11